Assignment 4

Q1 Solution
Let the stay time of each student follow an exponential distribution with mean

$$E[X] = 30 \text{ minutes}.$$

The expected total time can be computed using conditional expectation.

$$E[\text{Time}] = \big(5 + E[\text{stay of 2nd student}]\big)P(X \le 5)$$
$$+\big(E[X \mid X \ge 5] + E[\text{stay of 2nd student}]\big)P(X > 5).$$

Since

$$E[\text{stay of 2nd student}] = 30,$$

and using the memoryless property of the exponential distribution,

$$E[X \mid X \ge 5] = 5 + E[X] = 5 + 30 = 35.$$

Also,

$$P(X \le 5) = 1 - e^{-5/30},$$

and

$$P(X > 5) = e^{-5/30}.$$

Substituting these values:

$$E[\text{Time}] = (5 + 30)(1 - e^{-5/30}) + (35 + 30)e^{-5/30}.$$

Simplifying,

$$E[\text{Time}] = 35(1 - e^{-5/30}) + 65e^{-5/30},$$

$$= 35 - 35e^{-5/30} + 65e^{-5/30},$$

$$= 35 + 30e^{-5/30}.$$

Therefore,

$$E[\text{Time}] \approx 60.394 \text{ minutes}.$$

Q2 Solution
Uniform over disk $\Rightarrow$ area argument.

(a)

$$\mathbb{P}(X \le x) = \frac{\pi x^2}{\pi r^2} = \frac{x^2}{r^2}$$

Differentiate:

$$f_X(x) = \frac{2x}{r^2}, \quad 0 \le x \le r$$

(b)

$$\mathbb{E}[X] = \int_0^r x \frac{2x}{r^2} dx = \frac{2}{r^2} \int_0^r x^2 dx = \frac{2}{r^2} \cdot \frac{r^3}{3} = \frac{2r}{3}$$

(c)

$$\mathbb{E}[X^2] = \int_0^r x^2 \frac{2x}{r^2} dx = \frac{2}{r^2} \cdot \frac{r^4}{4} = \frac{r^2}{2}$$

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{r^2}{2} - \frac{4r^2}{9} = \frac{r^2}{18}$$

Q3 Solution

(a)

$$\mathbb{P}(H) = \mathbb{E}[P] = \int_0^1 p \cdot 6p(1-p) dp$$

$$= 6 \int_0^1 (p^2 - p^3) dp = 6 \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{1}{2}$$

(b)

$$f_{P|H}(p) = \frac{p f_P(p)}{\mathbb{P}(H)} = \frac{6p^2(1-p)}{1/2} = 12p^2(1-p)$$

(c)

$$\mathbb{E}[P|H] = \int_0^1 p \cdot 12p^2(1-p) dp$$

$$= 12 \int_0^1 (p^3 - p^4) dp = 12 \left(\frac{1}{4} - \frac{1}{5} \right) = \frac{3}{5}$$

Q4 Solution
Mixture distribution.

(a)

$$f_X(x) = 0.02\delta(x) + 0.98 \cdot 2e^{-2x}, \quad x > 0$$

(b)

$$\mathbb{P}(X \geq 1) = 0.98 \int_1^\infty 2e^{-2x} dx = 0.98e^{-2}$$

(c)

$$\mathbb{P}(X > 2 | X \geq 1) = \frac{e^{-4}}{e^{-2}} = e^{-2}$$

(d)

$$\mathbb{E}[X] = 0.02(0) + 0.98 \cdot \frac{1}{2} = 0.49$$

$$\mathbb{E}[X^2] = 0.98 \cdot \frac{2}{4} = 0.49$$

$$\text{Var}(X) = 0.49 - (0.49)^2 = 0.2499$$

Q5 Solution

(a)

$$F_X(x) = \int_{x_m}^{x} \frac{\alpha x_m^\alpha}{t^{\alpha+1}} dt = 1 - \left(\frac{x_m}{x}\right)^\alpha$$

(b)

$$\mathbb{P}(X > 3x_m | X > 2x_m) = \frac{(3x_m)^{-\alpha}}{(2x_m)^{-\alpha}} = \left(\frac{2}{3}\right)^\alpha$$

(c)

$$\mathbb{E}[X] = \frac{\alpha x_m}{\alpha - 1}, \quad \text{Var}(X) = \frac{\alpha x_m^2}{(\alpha - 1)^2(\alpha - 2)}$$

Solution to Question 6

Metro trains arrive every 15 minutes:

$$6:00,\ 6:15,\ 6:30,\ \ldots$$

You arrive between 7 : 10 and 7 : 30 a.m.
Let

$$X = \text{elapsed time after 7:10 when you arrive.}$$

Hence,

$$0 \le X \le 20.$$

Assuming your arrival time is uniformly distributed,

$$X \sim U(0, 20).$$

Therefore,

$$f_X(x) = \frac{1}{20}, \qquad 0 \le x \le 20.$$

The trains arrive at:

$$7:15 \quad \text{and} \quad 7:30.$$

The waiting time Y is:

$$Y = \begin{cases} 5 - X, & 0 \le X \le 5, \\ 20 - X, & 5 < X \le 20. \end{cases}$$

(a) CDF of Y
We compute:

$$F_Y(y) = P(Y \le y).$$

Since the maximum waiting time is 15 minutes,

$$0 \le y \le 15.$$

Now,

$$Y \le y$$

means:

$$5 - X \le y \quad \text{or} \quad 20 - X \le y.$$

Thus,

$$X \ge 5 - y \quad \text{or} \quad X \ge 20 - y.$$

Using the uniform distribution of X:
For $0 \le y \le 5$,

$$F_Y(y) = P(5 - y \le X \le 5) + P(20 - y \le X \le 20).$$

Each interval has length y, hence

$$F_Y(y) = \frac{y + y}{20} = \frac{y}{10}.$$

For $5 < y \le 15$, the first interval contributes completely:

$$P(0 \le X \le 5) = \frac{5}{20} = \frac{1}{4}.$$

The second interval contributes:

$$\frac{y}{20}.$$

Hence,

$$F_Y(y) = \frac{1}{4} + \frac{y}{20}.$$

Therefore,

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ \dfrac{y}{10}, & 0 \le y \le 5, \\ \dfrac{1}{4} + \dfrac{y}{20}, & 5 < y \le 15, \\ 1, & y > 15. \end{cases}$$

(b) PDF of Y
Differentiate the CDF:

$$f_Y(y) = \frac{d}{dy} F_Y(y).$$

Thus,

$$f_Y(y) = \begin{cases} \frac{1}{10}, & 0 \le y \le 5, \\ \frac{1}{20}, & 5 < y \le 15, \\ 0, & \text{otherwise.} \end{cases}$$

Solution to Question 7

The triangle has vertices:

$$(0,1), \quad (0,-1), \quad (1,0).$$

Area of the triangle:

$$A = \frac{1}{2}(2)(1) = 1.$$

Since the distribution is uniform,

$$f_{X,Y}(x,y) = 1$$

inside the triangle.

Define

$$Z = |X - Y|.$$

CDF of Z

We compute:

$$F_Z(z) = P(|X - Y| \le z).$$

The condition

$$|X - Y| \le z$$

means

$$-z \le X - Y \le z.$$

Equivalently,

$$X - z \le Y \le X + z.$$

The required probability equals the area of the region satisfying this condition inside the triangle.

After geometric computation,

$$F_Z(z) = \begin{cases} 0, & z < 0, \\ 2z - z^2, & 0 \le z \le 1, \\ 1, & z > 1. \end{cases}$$

PDF of Z

Differentiate:

$$f_Z(z) = \frac{d}{dz} F_Z(z).$$

Hence,

$$f_Z(z) = \begin{cases} 2 - 2z, & 0 \le z \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Solution to Question 8
Let

$$X, Y \sim U(0,1)$$

independently.
We compute:

$$E[|X - Y|] = \int_0^1 \int_0^1 |x - y| \, dx \, dy.$$

Using symmetry:

$$E[|X - Y|] = 2 \int_0^1 \int_0^x (x - y) \, dy \, dx.$$

Evaluate the inner integral:

$$\int_0^x (x - y) \, dy = \left[xy - \frac{y^2}{2} \right]_0^x = x^2 - \frac{x^2}{2} = \frac{x^2}{2}.$$

Therefore,

$$E[|X - Y|] = 2 \int_0^1 \frac{x^2}{2} \, dx.$$

$$= \int_0^1 x^2 \, dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}.$$

Hence,

$$\boxed{E[|X - Y|] = \frac{1}{3}}$$

Solution to Question 9
Initially the stick length is 1.
First break:

$$X \sim U(0,1).$$

Conditioned on $X = x$, the second break is uniform on $[0, x]$.
Hence:

$$Y | X = x \sim U(0, x).$$

Therefore,

$$f_{Y|X}(y|x) = \frac{1}{x}, \qquad 0 < y < x < 1.$$

Also,

$$f_X(x) = 1, \qquad 0 < x < 1.$$

(a) Joint PDF
Using

$$f_{X,Y}(x, y) = f_{Y|X}(y|x) f_X(x),$$

we obtain:

$$f_{X,Y}(x, y) = \frac{1}{x}, \qquad 0 < y < x < 1.$$

Otherwise,

$$f_{X,Y}(x, y) = 0.$$

(b) Marginal PDF of Y
Integrate over x:

$$f_Y(y) = \int_y^1 \frac{1}{x}\, dx.$$

$$= [\ln x]_y^1 = -\ln y, \qquad 0 < y < 1.$$

Hence,

$$f_Y(y) = \begin{cases} -\ln y, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(c) Compute $E[Y]$ using the PDF

$$E[Y] = \int_0^1 y(-\ln y)\, dy.$$

Using integration by parts:

$$u = -\ln y, \qquad dv = y\, dy.$$

Then,

$$du = -\frac{1}{y}\, dy, \qquad v = \frac{y^2}{2}.$$

Thus,

$$E[Y] = \left[-\frac{y^2}{2} \ln y\right]_0^1 + \frac{1}{2} \int_0^1 y\, dy.$$

$$= \frac{1}{2} \left[\frac{y^2}{2}\right]_0^1 = \frac{1}{4}.$$

Hence,

$$\boxed{E[Y] = \frac{1}{4}}$$

(d) Compute $E[Y]$ using $Y = X\left(\frac{Y}{X}\right)$
Observe that:

$$X \sim U(0, 1), \qquad \frac{Y}{X} \sim U(0, 1),$$

and they are independent.

Therefore,

$$E[Y] = E\left[X\left(\frac{Y}{X}\right)\right].$$

Using independence:

$$E[Y] = E[X]\, E\left[\frac{Y}{X}\right].$$

Since

$$E[X] = \frac{1}{2}, \qquad E\left[\frac{Y}{X}\right] = \frac{1}{2},$$

we obtain:

$$E[Y] = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Hence,

$$\boxed{E[Y] = \frac{1}{4}}$$